Geophys. J. Int. (1995) **122,** 227–242

Secondary palaeoremanence of Tethyan sediments from the Zanskar Range (NW Himalaya)

E. Appel,[1] A. Patzelt[1] and C. Chouker[2]

[1]*Institut für Geologie und Paläontologie, Sigwartstr.* 10, 72076 *Tübingen, Germany*
[2]*Institut für Geophysik, Theresienstr.* 41,80333 *München, Germany*

Accepted 1995 January 16. Received 1994 December 9; in original form 1994 May 6

Downloaded from https://academic.oup.com/gji/article/122/1/227/577748 by Curtin University Library user on 25 May 2026

SUMMARY

Palaeomagnetic investigations have been carried out on Tethyan sediments from the NW Zanskar Range (33.9°N, 76.5°E). A total of 455 oriented samples from 41 sites were drilled from four stratigraphic units of Middle Cretaceous to Early Eocene age (Shillakong Formation, Marpo Limestone/Spanboth Formation, Stumpata Quartzarenite/Spanboth Formation, Kong Formation).

A characteristic remanence (ChRM) could be isolated for 27 sites (254 specimens) through detailed thermal and alternating field demagnetization and multicomponent analysis. All the ChRM directions identified are similar, irrespective of geological age. They fail the fold test and yield an overall *in situ* mean direction of $D = 336.6°/I = +39.6°$ with $\alpha_{95} = 5.9°$ and $k = 22.9$. This corresponds to a palaeopole position at 66.5°N, 323.5°E (N-pole). Isothermal remanence (IRM) acquisition and thermal demagnetization of a saturation IRM identify pyrrhotite as the dominant ferrimagnetic mineral and carrier of the ChRM. The pyrrhotite remanence is probably a thermoremanent magnetization acquired when the temperature of low-grade metamorphism decreased below about 300°C.

The ChRM directions are thus clearly secondary in origin, and a negative conglomerate test carried out on the Kong Formation demonstrates that remanence acquisition post-dates the Early Eocene. The mean ChRM inclination suggests that the remanence was acquired at about 20°N. Cooling events in the higher Himalayan crystallines indicate an Oligocene age for this remanence acquisition. The mean ChRM declination shows a counterclockwise rotation of 23.4° since remanence acquisition. Assuming remanence acquisition at about 30 Myr ago, a slight counterclockwise rotation of about 10° relative to the stable Indian Plate can be concluded. This does not fit with the general pattern of palaeomagnetic results from neighbouring areas within the western syntaxis of the Himalaya, from which a clockwise rotation relative to the stable Indian Plate is expected for the Zanskar Range.

Key words: block rotation, palaeomagnetism, pyrrhotite, Tethyan Himalaya, Zanskar.

INTRODUCTION

In the Early Cretaceous, India separated from Gondwana and drifted northwards with a maximum drift rate of up to 18–20 cm yr^{-1} during Maastrichtian to Early Tertiary times (Patriat & Achache 1984; Klootwijk *et al.* 1991). A drop in drift rate to less than 5 cm yr^{-1} at about 50–55 Myr (Patriat & Achache 1984; Besse & Courtillot 1988) is usually interpreted as the beginning of the India–Eurasian collision. However, during the last few years an initial contact of India with Eurasia at around the Cretaceous–Tertiary time boundary has been discussed (Jaeger, Courtillot & Tapponnier 1989; Klootwijk *et al.* 1991; Tonarini *et al.* 1993; Patzelt *et al.* 1995). The collision started in the western part with progressive closure of the Neo-Tethyan ocean from west to east (Klootwijk, Conaghan & Powell 1985). Ongoing convergence between the Indian and Eurasian plates led to the formation of the western Himalayan syntaxis, oroclinal

bending of the Himalaya, rise of the Tibetan Plateau and crustal shortening within north–south-propagating thrust zones (Klootwijk *et al.* 1985, 1986; Powell 1986; Appel & Li 1988; Dewey, Cande & Pitman 1989). The Himalayan zone can be subdivided into several tectonostratigraphic units (Gansser 1964; Fuchs 1982a; Gaetani & Garzanti 1991). From north to south these are: a Precambrian–Eocene sedimentary succession deposited on the passive northern margin of India (Tethyan Himalaya), now situated directly south of the ophiolites of the Indus–Yarlung suture zone, the High Himalayan crystalline zone (paragneisses and Tertiary granites), the Lesser Himalaya south of the Main Central Thrust (MCT), the Neogene molasse belt (Siwaliks) south of the Main Boundary Thrust (MBT) and Quaternary sediments south of the Main Frontal Thrust (Fig. 1).

Palaeomagnetic data from the Tethyan Himalaya at the leading edge of the Indian Plate during its Late Cretaceous and Cainozoic northward drift could potentially provide information about the former northern extent of India and the geodynamic evolution of the convergence zone. However, very few palaeomagnetic data are available for the Tethyan Himalaya (Klootwijk & Bingham 1980; Klootwijk *et al.* 1983; Besse *et al.* 1984; Zhu & Teng 1984; Appel, Müller & Widder 1991), mainly due to the relative inaccessibility of this zone for logistical and political reasons, and due to widespread metamorphic overprints.

In this paper we present the results of a palaeomagnetic investigation carried out in the Tethyan Himalaya of the Zanskar Range, NW India (Fig. 1).

GEOLOGY AND SAMPLING

The Cretaceous–Eocene succession of the Zanskar Range is strongly folded, and shows a great variety of sedimentary environments (Fuchs 1987; Fuchs & Willems 1990). The Shillakong Formation, deposited from Turonian to Early Campanian times consists of multicoloured pelagic limestones of a couches rouges facies. The younger Spanboth Formation can be subdivided into three members, which some authors (e.g. Gaetani & Garzanti 1991) classify as separate formations: the Marpo Limestone comprising Late Maastrichtian shallow-water platform carbonates, the Stumpata Quartzarenite representing a coastal sandstone package deposited at the Cretaceous–Tertiary time boundary, and the Dibling/Lingshet Limestone consisting of Late Palaeocene open-shelf carbonates (Gaetani *et al.* 1983; Fuchs 1987; Fuchs & Willems 1990). The Early Eocene Kong Formation, composed mainly of slates and carbonates, forms the youngest marine sedimentary unit of the Zanskar Range and is found only around the Spongtang Klippe. With the deposition of the Chulung La Red Beds (Eocene), sedimentation in the Tethyan Himalaya of Zanskar ended. The main folding and metamorphism of the Zanskar shelf was initiated when the marine sedimentation in Zanskar was finished (Fuchs & Willems 1990).

Figure 2 shows a geologic map of the area investigated (Fuchs 1982b) with site locations indicated. Sampling was carried out during two field campaigns in the summer seasons of 1988 and 1990, using a portable rock drill.

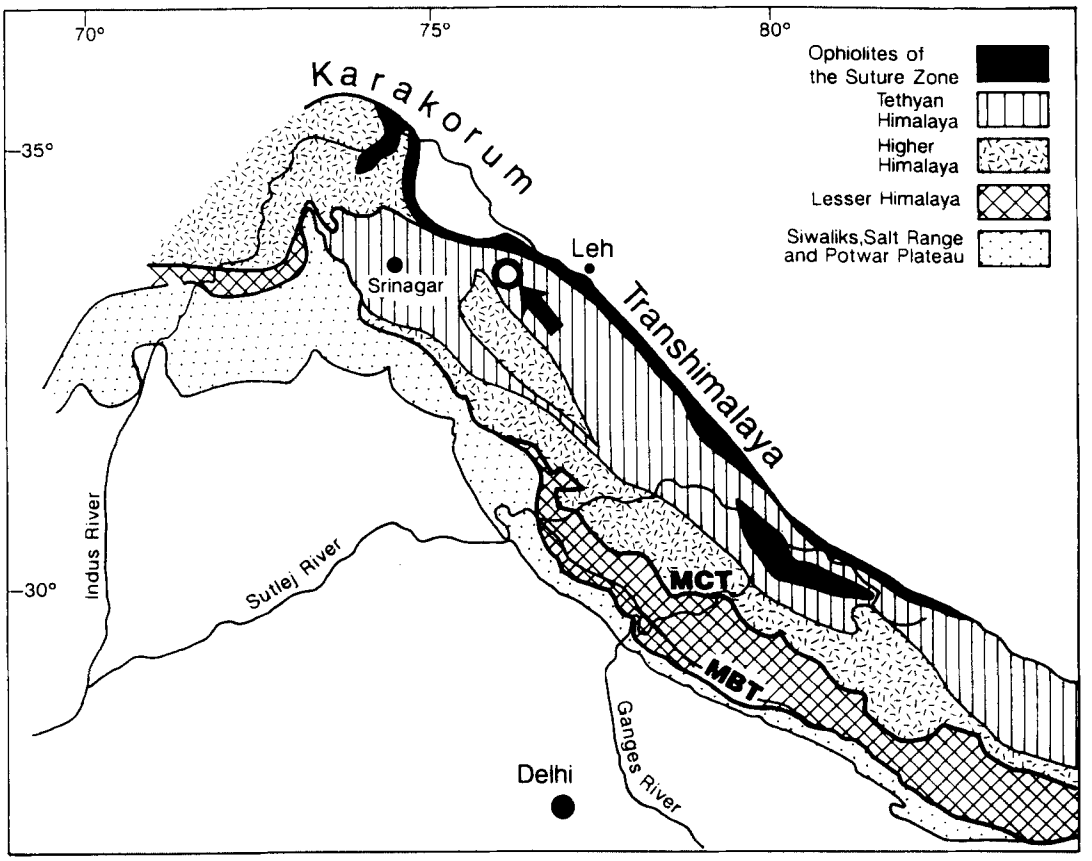


Figure 1. Tectonostratigraphic map of the western part of the Himalaya (simplified after Gansser 1964) showing the location of the investigation area (circle and arrow).

Downloaded from https://academic.oup.com/gji/article/122/1/227/577748 by Curtin University Library user on 25 May 2026

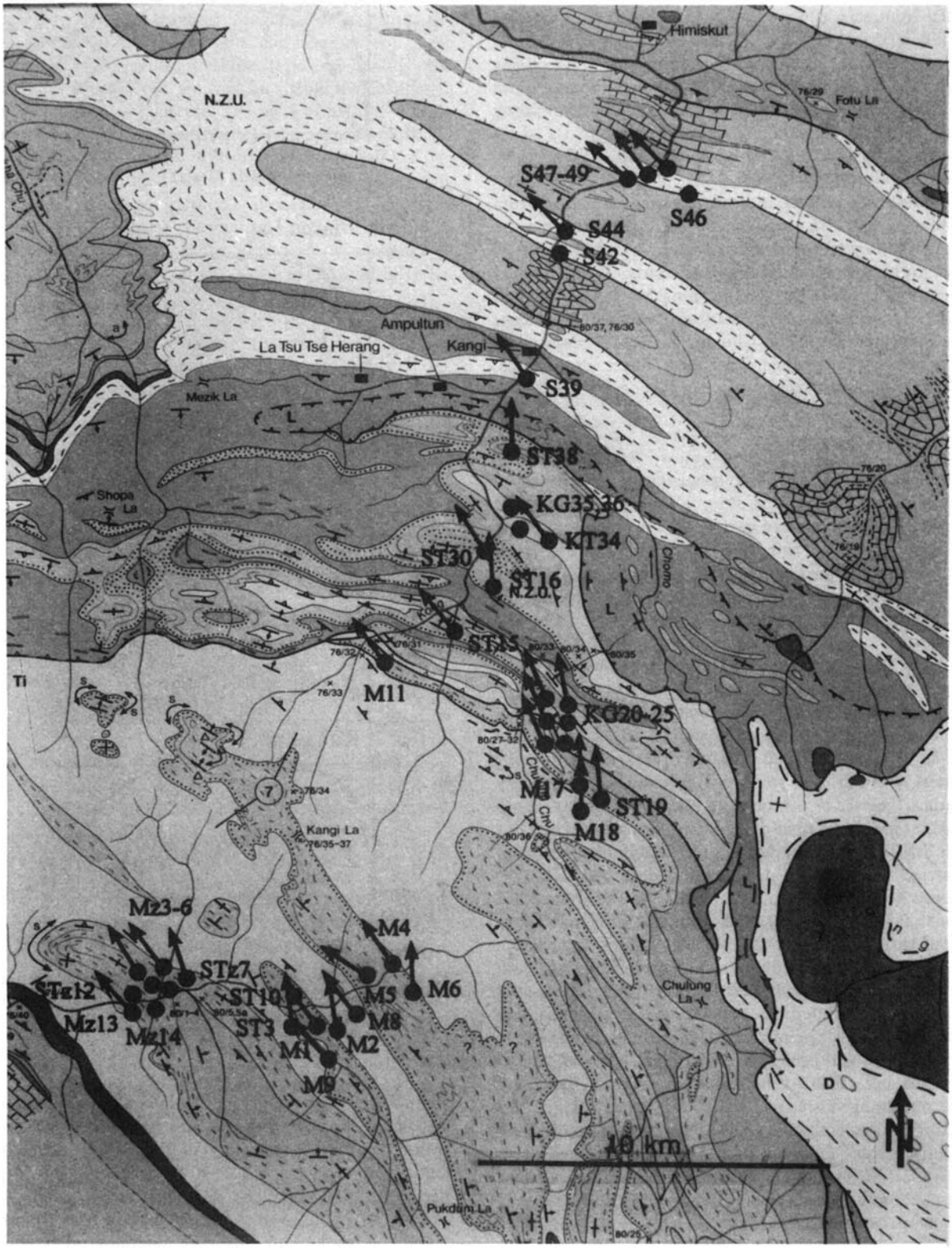


Figure 2. Geological map of the western Zanskar Range (Fuchs 1982b). The site locations are marked by black dots (S, Shillakong Formation; M, Mz, Marpo Limestone; ST, STz, Stumpata Quartzarenite; KG, Kong Formation; KT, conglomerate of Kong Formation). Formations can be identified by the site locations. *In situ* declination values for the ChRM1 site mean directions are shown by arrows. Coordinates of the village of Himiskut (upper edge of the map) are 34°18′N, 76°38′E.

Downloaded from https://academic.oup.com/gji/article/122/1/227/577748 by Curtin University Library user on 25 May 2026

Orientation of the cores was carried out with a magnetic compass. Usually, 10 samples of 2.5 cm diameter were collected from each site. In total, 455 oriented core samples were taken from 41 sites in the Shillakong Formation, Marpo Limestone, Stumpata Quartzarenite and Kong Formation. One site was drilled in a conglomerate, from clasts derived from debris flow material of the Kong Formation. The conglomerate was formed only a short time after deposition of the Kong Formation (G. Fuchs, private communication, 1990).

In addition, more than 150 samples were collected from the Dibling/Lingshet Limestone, about 70 samples from the Kioto Formation (Rhaetian–Early Dogger) and about 30 samples from the Chulung La Red Beds. However, the NRM intensities from the Dibling/Lingshet Limestone and the Kioto Formation were too low to yield significant results. Specimens from the Chulung La Red Beds showed highly scattered remanence directions. These three units, therefore, are not discussed further in this paper.

SAMPLE TREATMENT

The core samples were cut into specimens with a length of 2.2–2.3 cm. Detailed stepwise thermal and alternating field (AF) demagnetizations were performed for one pilot sample (mostly twin specimens from one core) from each site. After each thermal demagnetization step, the bulk susceptibility was measured in order to control major chemical changes in the magnetic mineral composition. One specimen per site was subjected to isothermal remanence acquisition (IRM) in a DC field up to 1.5 T with subsequent stepwise thermal demagnetization of the saturation IRM (SIRM), in order to obtain information about the magnetic mineralogy. Depending on the results of the pilot studies, IRM acquisition curves and SIRM thermal decay curves, the remaining specimens of each site were demagnetized thermally, in alternating fields, or through a combination of both methods. Stepwise thermal demagnetization was carried out with an MMTD1-furnace. Alternating field demagnetization was carried out with a μ-metal shielded 2G-degausser, which was hosted in a two-pair Helmholtz coil system for additional field compensation. Remanence directions were measured with a cryogenic magnetometer (Remánometre á Squid RS 01, LET1 Grenoble) with a noise level of about 0.02 mA m^{-1}. Orthogonal vector plots and equal-area projections were plotted to examine the demagnetization behaviour of the specimens. Magnetic directions were determined through principal component analysis (PCA) after Kirschvink (1980). Calculations of mean values were carried out using Fisherian statistics (Fisher 1953). The anisotropy of susceptibility (AMS) was also measured for characteristic sites, in order to detect possible correlations between the rock magnetic fabric and remanence directions. All susceptibility measurements were carried out with a KLY-2 kappabridge.

MAGNETIC MINERALOGY

IRM-acquisition curves and thermal demagnetization of SIRMs demonstrate the similarity of the magnetomineralogical compositions of all four stratigraphic units studied (Fig. 3). Specimens of the Kong Formation, the

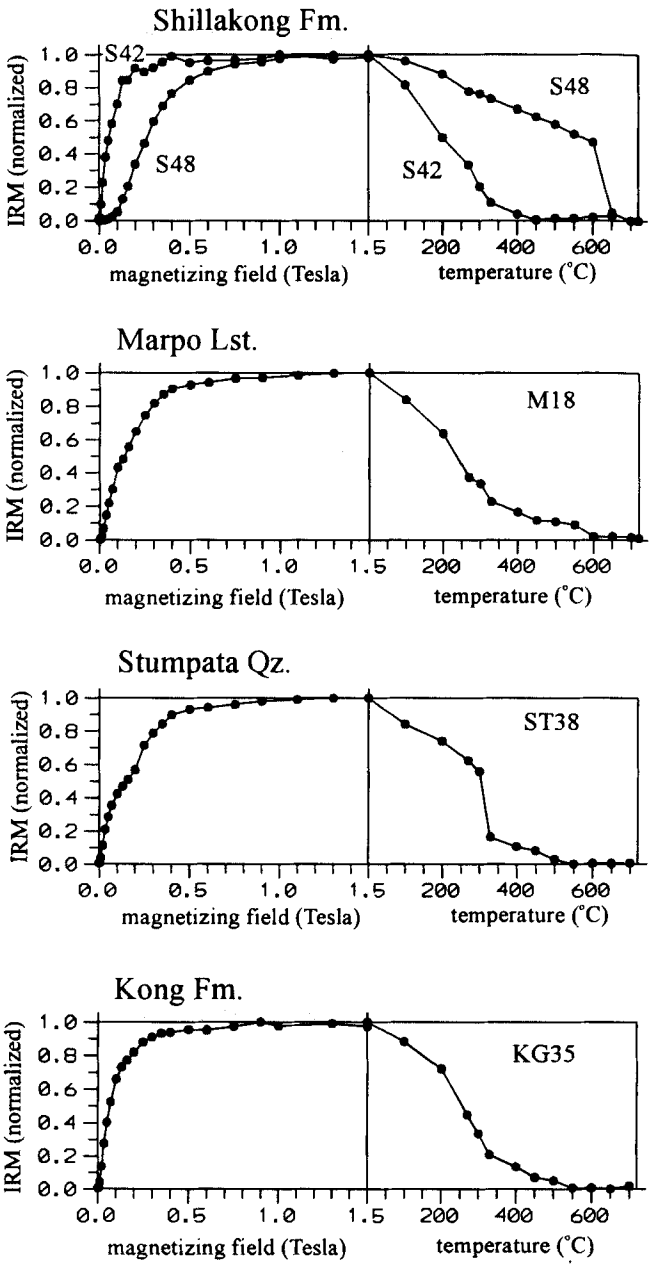


Figure 3. Acquisition curves and thermal demagnetization of isothermal remanent magnetization for representative specimens of the stratigraphic units investigated.

Marpo Limestone and the Stumpata Quartzarenite show remanence carriers with low and medium coercivity, with a major decrease in intensity of the SIRM during thermal demagnetization at 300°–320°C, followed by a gradual decrease until complete demagnetization at 550°–600°C. Specimens of the Shillakong Formation show a somewhat greater variation in magnetomineralogical composition, varying from specimens with a low to medium coercivity and a notable decrease of magnetization intensity at 320°C (S42 in Fig. 3), to specimens with a predominant component of high coercivity and a major decrease of the SIRM intensity at about 650°C, without a major decrease in intensity at 300°C (S48 in Fig. 3).

Downloaded from https://academic.oup.com/gji/article/122/1/227/577748 by Curtin University Library user on 25 May 2026

Thus, for all stratigraphic units, pyrrhotite, with medium coercivity and a Curie-temperature of about 320°C, is the dominant ferrimagnetic mineral. Maghemite, which usually converts to haematite at 300°–400°C can be ruled out as the cause of the 300°C decay, because repeated heating and cooling cycles up to 400°C showed no significant changes in bulk susceptibility. Magnetite (Kong Formation, Marpo Limestone, Stumpata Quartzarenite) and haematite (Shillakong Formation, rarely in Marpo Limestone and Kong Formation) are identified additionally by their unblocking temperature spectra. Goethite also occurs in some specimens of the Kong Formation and the Stumpata Quartzarenite in small amounts, identified by a change in the thermal demagnetization curves of SIRM and NRM at 100°C.

NRM intensities varied from 0.05–1.5 mA m^{-1} (Shillakong Formation), 0.2–10.0 mA m^{-1} (Marpo Limestone) and a 0.1–9.0 mA m^{-1} (Stumpata Quartzarenite) to 0.02–1.0 mA m^{-1} (Kong Formation and conglomerate), with bulk susceptibilities of 20–200 $\times 10^{-6}$ SI (Shillakong Formation), 50–35 $\times 10^{-6}$ SI (Marpo Limestone), 0–100 $\times 10^{-6}$ SI (Stumpata Quartzarenite) and 0–300 $\times 10^{-6}$ SI (Kong Formation and conglomerate).

PALAEOMAGNETIC RESULTS

Shillakong Formation (seven sites) (Fig. 4, Table 1)

Thermal demagnetization was carried out, because of the coexistence of pyrrhotite and haematite. A remanence component carried by pyrrhotite was removed up to about 350°C (ChRM1). For five sites, a reliable mean direction for the ChRM1 could be identified by PCA between 280° and 320°C. Two sites (S42, S46) produced no interpretable results because the pyrrhotite contribution was too low. Above 350°C a stable direction was revealed in some of the specimens (haematite component: ChRM2) with about the same direction as the pyrrhotite component. Owing to thermochemical alteration (above 450°–500°C) and very low intensity values, this ChRM2 could be reliably identified only in site S44.

Marpo Limestone Member/Spanboth Formation (16 sites) (Fig. 5, Table 2)

The specimens were first AF-treated in order to demagnetize remanences carried by magnetite. After

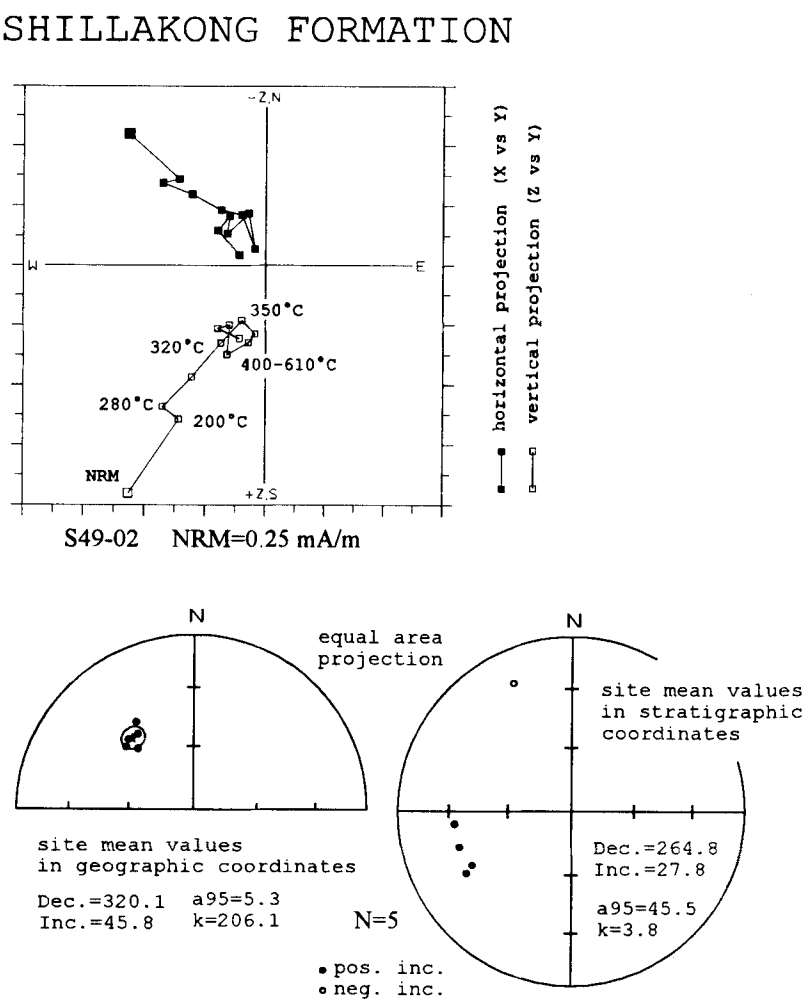


Figure 4. Top: orthogonal vector plot showing the demagnetization behaviour of a representative specimen from the Shillakong Formation. Bottom: site mean directions of the ChRM1 component (equal area projection) before (left side) and after (right side) tectonic correction. The α_{95}-circle for the formation mean direction is shown for the geographic coordinates.

Downloaded from https://academic.oup.com/gji/article/122/1/227/577748 by Curtin University Library user on 25 May 2026

Table 1. Statistical parameters for the Shillakong Formation.

SHILLAKONG FORMATION

				ChRM1 (pyrrhotite component)					
site	N	spec.	bedding	geographic corr.				stratigraphic corr.	
name		n / r		Dec.	Inc.	a95	k	Dec.	Inc.
S39	8	7 / 0	210 / 71	327.1	40.0	17.5	12.9	263.8	33.0
S42	5	–	206 / 82	limited pyrrhotite content					
S44	9	5 / 0	15 / 87	317.5	44.8	26.5	9.3	335.5	−20.3
S46	4	–	193 / 75	limited pyrrhotite content					
S47	10	6 / 0	195 / 74	313.3	46.7	25.8	7.7	239.4	30.8
S48	10	9 / 0	205 / 71	323.9	45.7	12.8	45.7	252.3	33.1
S49	9	8 / 0	199 / 67	317.9	51.4	15.5	13.7	241.5	35.5
		SHILLAKONG (total)		**320.1 a95=5.3**	**45.8 k=206.1**		**5 sites**	**233.1 a95=37.5**	**36.9 k=5.1**

				ChRM2 (hematite component)					
site	N	spec.	bedding	geographic corr.				stratigraphic corr.	
name		n / r		Dec.	Inc.	a95	k	Dec.	Inc.
S44	9	6 / 0	15 / 87	335.3	54.0	18.4	14.2	350.7	−24.5

N, total number of measured specimens; *n*/*r*, number of normal/reverse polarity specimen directions used for calculation of the site mean direction; bedding, azimuth and dip of the dip arrow. Normal (reverse) directions are shown when $n > r$ ($r > n$).

destruction of a very soft magnetization (multidomain magnetite?) a component representing the present field was demagnetized in the range between 10/15 and 30/50 mT. The direction and intensity values of the residual magnetization above 30/50 mT (ChRM1) remained stable up to 100 mT. Subsequent thermal demagnetization revealed unblocking temperatures of about 300°C indicating pyrrhotite as the carrier of remanence. The ChRM1 could be reliably identified in 13 sites. Both normal and reverse polarities were observed. However, two sites (M4, M5 from a overturned fold limb) show mean directions strongly deviating from the other 11 sites. At least for site M5, haematite was identified as an additional ferrimagnetic mineral by IRM curves. Possibly pyrrhotite and haematite components are superimposed in these sites. Thus M4 and M5 were not used for calculating the formation mean direction. For three other sites (Mz4, Mz6, Mz14) a different demagnetization procedure was selected (Mz4, Mz6: AF 0–50 mT, then thermal 0°–300°C; Mz14: thermal 0°–200°C, then AF 0–30 mT), which turned out to be unsuitable for separating the pyrrhotite component (haematite was also observed in Mz4 and Mz6, and goethite was occasionally found to occur in Mz14).

Stumpata Quartzarenite Member/Spanboth Formation (9 sites) (Fig. 6, Table 3)

The specimens were first heated to 200°C in order to remove the sporadic occurrence of an overprint residing in goethite. With subsequent AF-demagnetization, remanences carried by magnetite were removed. Directions of the magnetite component proved to be strongly scattered. The residual magnetization above 30 mT remained stable in alternating fields up to 100 mT and was demagnetized through a subsequent additional thermal treatment between 280° and 320°C (ChRM1, carried by pyrrhotite). Normal and reverse polarities occur. Seven sites were used for calculation of the formation mean direction; however, most of these show rather low values of *k*. Two further sites (ST10, STz12), with values of $k < 3$, were not included.

Kong Formation (eight sites) (Fig. 7, Table 4)

Because of the occasional occurrence of goethite or haematite (besides predominant pyrrhotite and magnetite), thermal demagnetization was carried out. The component demagnetized between 200° and 320°C (related to pyrrhotite) was used to calculate the ChRM1. Significant results could be obtained for four sites. Four other sites (KG21, KG22, KG35, KG36) showed highly scattered directions and were not analysed further.

Fold tests

The ChRM1 direction fails to pass the fold test of McElhinny (1964) at the 99 per cent confidence level (Shillakong Formation, Stumpata Quartzarenite, all sites

Downloaded from https://academic.oup.com/gji/article/122/1/227/577748 by Curtin University Library user on 25 May 2026

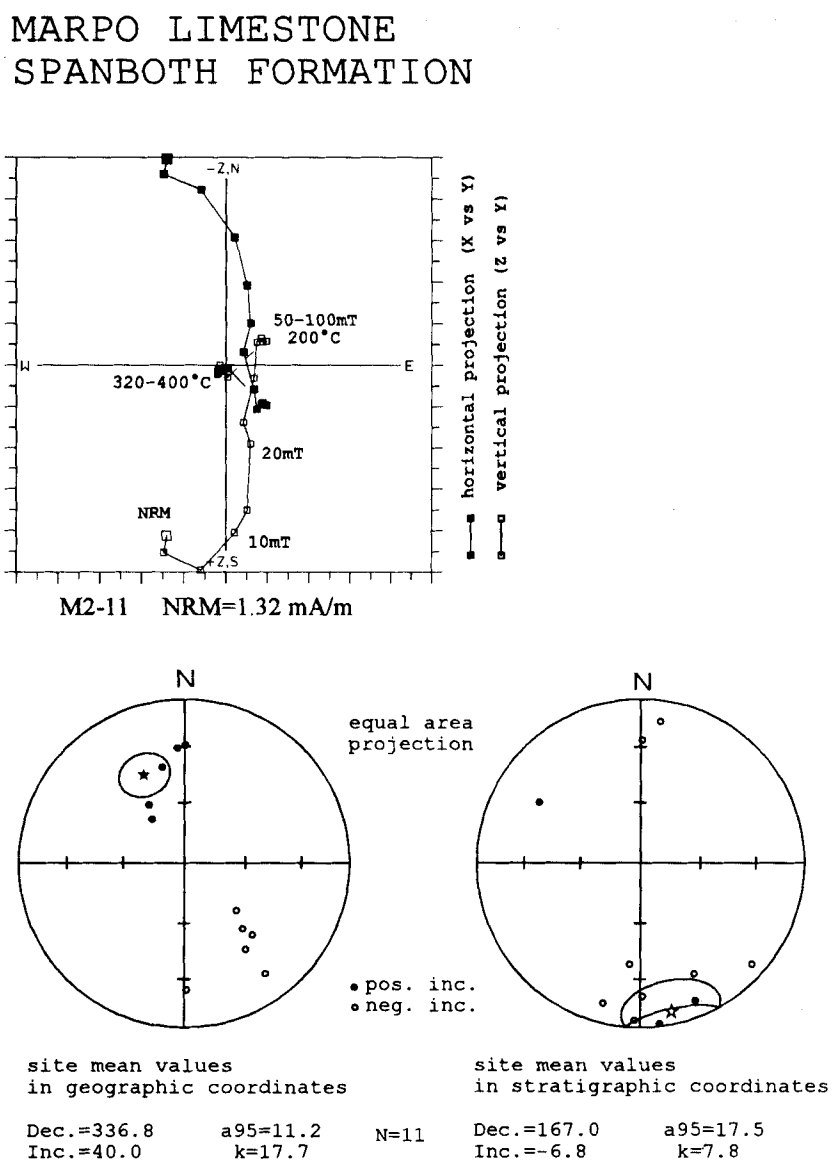


Figure 5. Top: orthogonal vector plot showing the demagnetization behaviour of a representative specimen from the Marpo Limestone. Bottom: site mean directions of the ChRM1 component (equal area projection) before (left side) and after (right side) tectonic correction. The formation mean direction is shown by a star together with the α_{95}-circle before and after tectonic correction.

together), the '95 per cent level (Marpo Limestone), or somewhat below the 95 per cent level (Kong Formation). Clearly the ChRM1 directions represent a post-folding secondary remanence.

Conglomerate test (Fig. 8, Table 4)

Contrary to most source rock sites only magnetite and pyrrhotite, but no signs of goethite or haematite, were found in the conglomerate. Thus AF-demagnetization was used to remove a magnetite component (ChRM-A). A second component remains stable above 50 mT and shows a typical breakdown around 300°C during subsequent thermal treatment (ChRM-B, carried by pyrrhotite). The ChRM-A direction corresponds to the present magnetic field, whereas the ChRM-B is identical with the ChRM1 of the Kong Formation. The negative conglomerate test again demonstrates the secondary nature of the ChRM-B/ChRM1 direction. The formation age of the conglomerate indicates that this component was acquired some time after the Early Eocene.

AMS measurements

The anisotropy of susceptibility (AMS) was measured for representative specimens of each formation in order to see if there is any correlation between the rock magnetic fabric and the remanence directions that were identified. In Fig. 9 the degree of anisotropy (P-factor: k_{max}/k_{min}) is plotted versus the shape of the susceptibility tensor surface (E-factor: $k^2_{int}/(k_{max}\,k_{min})$). Only a very low degree of anisotropy was found for the Kong Formation and the Stumpata Quartzarenite. The Marpo Limestone shows a weak anisotropy, but the tensor shape is neither prolate nor oblate. The Shillakong Formation exhibits a strong degree of anisotropy with an oblate shape of the tensor surface. The

Downloaded from https://academic.oup.com/gji/article/122/1/227/577748 by Curtin University Library user on 25 May 2026

Table 2. Statistical parameters for the Marpo Limestone.

MARPO LIMESTONE / SPANBOTH FORMATION

				ChRM1 (pyrrhotite component)					
site	N	spec.	bedding	geographic corr.				stratigraphic corr.	
name		n / r		Dec.	Inc.	a95	k	Dec.	Inc.
M1	10	0 / 7	238 / 57	143.9	−17.8	10.0	37.5	131.7	−8.8
M2	10	5 / 3	243 / 59	351.0	38.8	20.6	8.2	303.9	32.2
M4 *	9	8 / 0	69 / 68	146.4	58.8	9.4	35.6	100.4	12.6
M5 *	10	10 / 0	66 / 69	123.4	57.6	26.8	4.2	93.2	1.8
M6	9	0 / 9	65 / 27	178.8	−24.0	6.6	61.9	186.4	11.3
M8	10	0 / 7	37 / 61	142.6	−44.0	11.6	27.9	172.2	−9.2
M9	10	0 / 8	27 / 28	137.4	−39.7	6.6	71.9	153.6	−25.9
M11	11	0 / 9	31 / 76	145.3	−35.8	8.7	32.1	158.4	10.8
M17	11	11 / 0	37 / 63	359.0	30.9	9.7	23.3	2.2	−22.1
M18	9	9 / 0	37 / 66	0.2	29.8	8.4	39.0	1.2	−25.6
Mz3	31	29 / 0	19 / 69	328.6	55.6	8.8	10.3	173.4	2.0
Mz4	6	–	9 / 68	limited pyrrhotite content					
Mz5	21	20 / 0	9 / 68	324.1	63.4	4.8	47.0	182.2	−5.4
Mz6	17	–	9 / 68	limited pyrrhotite content					
Mz13	16	0 / 12	63 / 48	138.6	−46.4	3.5	151.3	185.6	−37.8
Mz14	20	–	65 / 46	limited pyrrhotite content					
			MARPO (total)	**336.8 a95=11.2**	**40.0 k=17.7**	**11 sites**		**167.0 a95=17.5**	**−6.8 k=7.8**

* Not used for mean calculation (see Table 1 for further explanations).

axes of minimum susceptibility are normal to the bedding plane and the axes of maximum susceptibility show a clearly preferred direction within the bedding plane. However, the mean remanence direction of the Shillakong Formation does not lie on the great circle defined by the other formation mean directions and the maximum susceptibility axis (Fig. 10). Hence no influence of the magnetic fabric on the mean direction of the Shillakong Formation can be concluded from the AMS data.

DISCUSSION

For all stratigraphic units a characteristic remanence carried by pyrrhotite has been determined (ChRM1). Negative fold tests and a negative conglomerate test indicate a post-tectonic origin of remanence acquisition. The ChRM1 results (Table 5) for all formations are similar, independent of their geological age, suggesting an approximately identical age of remanence acquisition. The formation mean directions are shown in Fig. 10. For all 27 sites (254 specimens) an overall mean remanence direction is calculated (Fig. 11, Table 5). The ChRM1 probably is a thermoremanent magnetization (TRM). Multistage thrusting resulting from the India–Eurasia collision occurred in the Zanskar shelf sediments, accompanied by regional metamorphism. Peak metamorphism in the Zanskar Range was reached at the end of the Eocene. The degree of metamorphism in the Zanskar area was low grade with temperatures of about 300°C, and was even present in the upper sedimentary layers, as indicated by X-ray and petrographic analyses (Garzanti & Brignoli 1989). Minimum temperatures of 300°C were also obtained from illite crystallinity for samples from our investigation area (Pflästerer, private communication, 1992). This remanence can be regarded as a TRM of the pyrrhotite component which was acquired when low-grade metamorphism decreased below temperatures of about 300°C. The occurrence of secondary remanences carried by pyrrhotite in sediments of the Tethyan Himalaya has also been reported from the Manang area in north central Nepal (Appel *et al.* 1991).

Downloaded from https://academic.oup.com/gji/article/122/1/227/577748 by Curtin University Library user on 25 May 2026

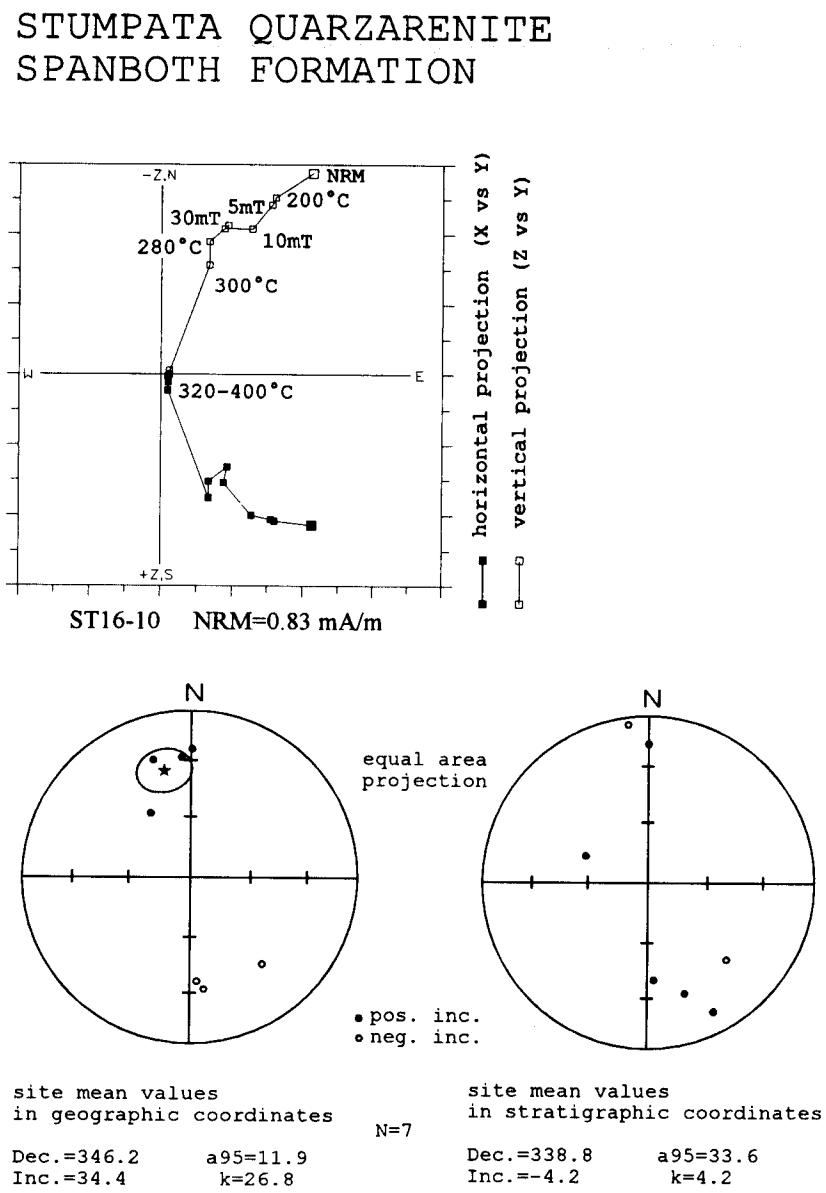


Figure 6. Top: orthogonal vector plot showing the demagnetization behaviour of a representative specimen from the Stumpata Quartzarenite. Bottom: site mean directions of the ChRM1 component (equal area projection) before (left side) and after (right side) tectonic correction. The formation direction is shown by a star together with the α_{95}-circle in geographic coordinates.

According to Rochette (1988), secondary formation of pyrrhotite from magnetite or from pyrite is to be expected in carbonates during low-grade metamorphism.

The ChRM1 mean inclination suggests that the remanence was acquired at a palaeolatitude of 22.5°N ± 4.2°. In Fig. 12 this result is compared with expected palaeolatitudes and palaeodeclinations for the sampling area according to the geological age of the formations studied, calculated from an apparent polar wander path (APWP) (Klootwijk *et al.* 1991) for India assuming no relative motion between the Zanskar Range and the Indian Shield. The older age limit of the remanence acquisition age is fixed by the age of the conglomerate of the Kong Formation (approximately identical to the age of the Kong Formation), whereas the younger age limit ranges up to the present within the calculated confidence ranges. However, there are several arguments to favour an acquisition age of not younger than about 25–30 Myr.

(1) The observed palaeolatitude (calculated from the ChRM1 inclination) should be further north than the expected one because of the north–south convergence of the Zanskar Range relative to stable India since remanence acquisition (e.g. Coward *et al.* 1986: 470 km shortening for the NW Himalaya, corresponding to about 4° for the palaeolatitude). The difference between the observed and expected palaeolatitudes could be additionally increased by inclination steepening through thrusting (northward tilting) along the MCT. Such an inclination steepening was observed by Rochette *et al.* (1994) and Appel *et al.* (1991) to account for 10° in the Everest area and 19° in the Manang area, respectively; however, the regional significance of this effect is not yet clear.

Downloaded from https://academic.oup.com/gji/article/122/1/227/577748 by Curtin University Library user on 25 May 2026

Table 3. Statistical parameters for the Stumpata Quartzarenite. (See Table 1 for further explanation.)

STUMPATA QUARTZARENITE / SPANBOTH FORMATION

site name	N	spec. n / r	bedding	ChRM1 (pyrrhotite component) geographic corr. Dec.	Inc.	a95	k	stratigraphic corr. Dec.	Inc.
ST3	7	0 / 7	242 / 54	173.0	−32.5	33.7	4.2	134.3	−33.9
ST10	9	–	47 / 23			no result (k<3)			
ST15	10	2 / 6	205 / 73	140.6	−32.7	9.4	35.7	152.7	13.8
ST16	10	2 / 4	354 / 78	176.2	−37.1	30.7	5.7	176.3	40.8
ST19	9	8 / 0	40 / 61	355.5	28.7	12.7	20.0	359.8	−18.0
ST30	9	7 / 0	21 / 71	328.4	52.9	29.2	5.2	352.5	−5.2
ST38	9	4 / 2	216 / 70	0.4	24.1	28.5	6.5	293.1	56.4
STz7	23	21 / 0	13 / 65	342.3	27.2	16.4	4.7	161.3	30.4
STz12	16	–	49 / 38			no result (k<3)			
		STUMPATA (total)		**346.2 a95=11.9**	**34.4 k=26.8**	**7 sites**		**338.8 a95=33.6**	**−4.2 k=4.2**

KONG FORMATION

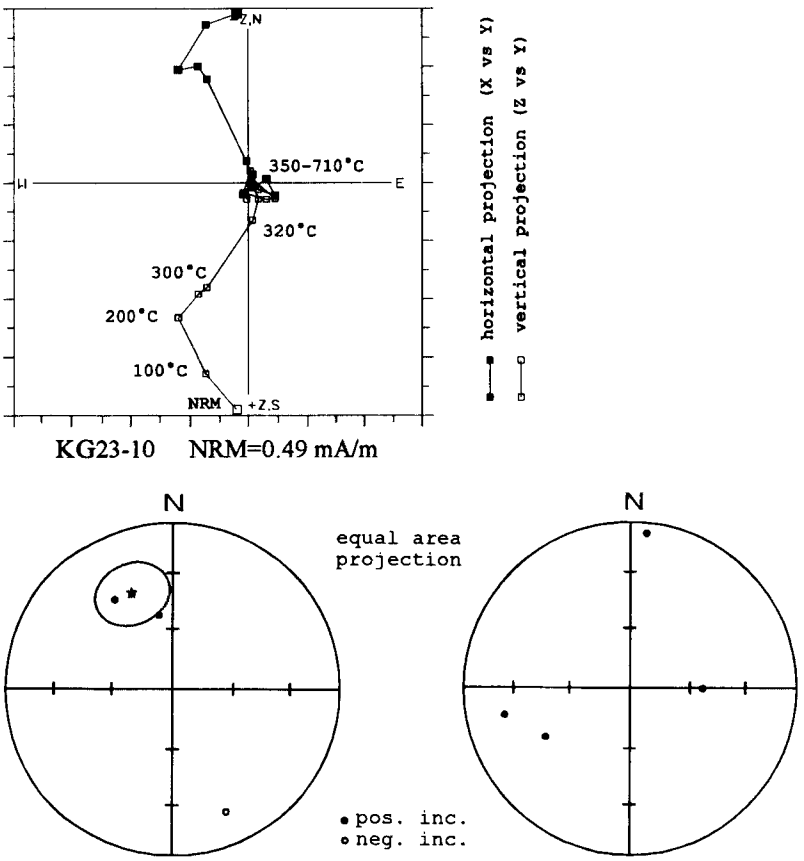


Figure 7. Top: orthogonal vector plot showing the demagnetization behaviour of a representative specimen from the Kong Formation. Bottom: site mean directions of the ChRM1 component (equal area projection) before (left side) and after (right side) tectonic correction. The formation mean direction is shown by a star together with the α_{95}-circle in geographic coordinates.

Downloaded from https://academic.oup.com/gji/article/122/1/227/577748 by Curtin University Library user on 25 May 2026

Table 4. Statistical parameters for the Kong Formation and conglomerate.

KONG FORMATION

				ChRM1 (pyrrhotite component)					
site	N	spec.	bedding	geographic corr.				stratigraphic corr.	
name		n / r		Dec.	Inc.	a95	k	Dec.	Inc.
KG20	6	0 / 5	202 / 66	156.6	−21.3	10.4	54.7	89.8	−54.0
KG21	12	–	195 / 75	strong scattering of specimen directions					
KG22	13	–	13 / 80	strong scattering of specimen directions					
KG23	16	10 / 1	208 / 89	327.1	36.5	9.0	26.5	258.1	23.7
KG24	4	3 / 1	197 / 84	337.2	38.3	29.4	10.7	239.6	41.1
KG25	9	7 / 0	29 / 52	349.7	53.0	14.7	17.7	6.2	7.2
KT34 *	14	0 / 10	–	145.4	−23.6	4.9	97.1	–	–
KG35	5	–	205 / 87	strong scattering of specimen directions					
KG36	9	–	200 / 80	strong scattering of specimen directions					
			KONG (total)	**336.8 a95=16.8**	**37.5 k=30.7**		**4 sites**	**235.2 a95=53.3**	**32.6 k=3.9**

* Not used for mean calculation (KT, conglomerate) (see Table 1 for further explanation).

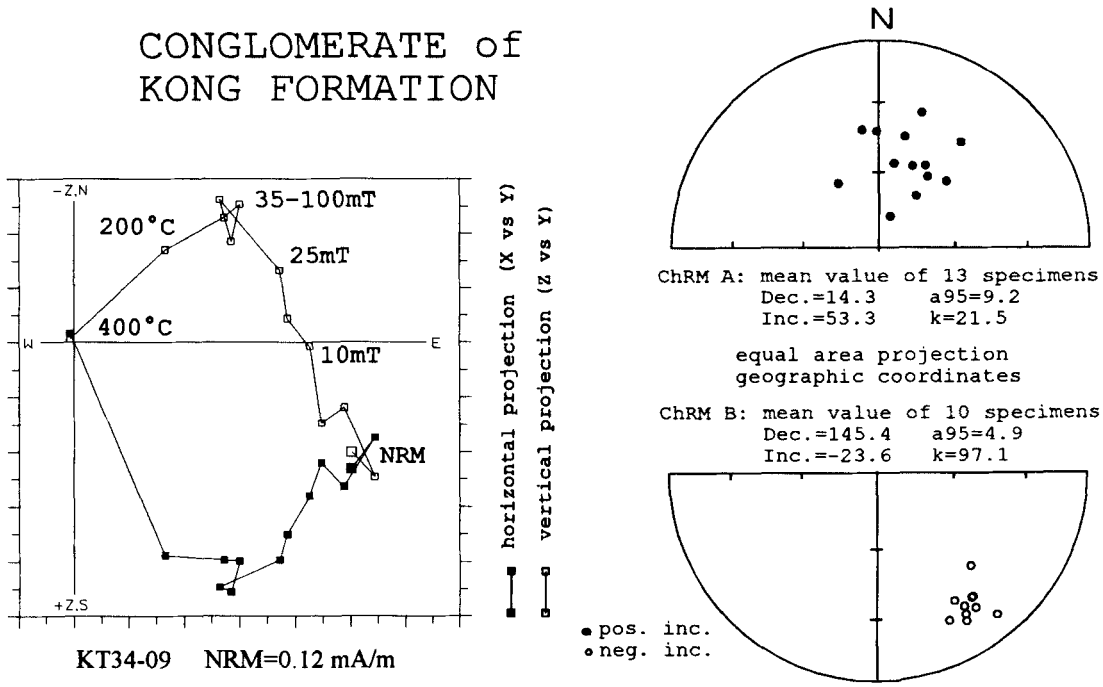
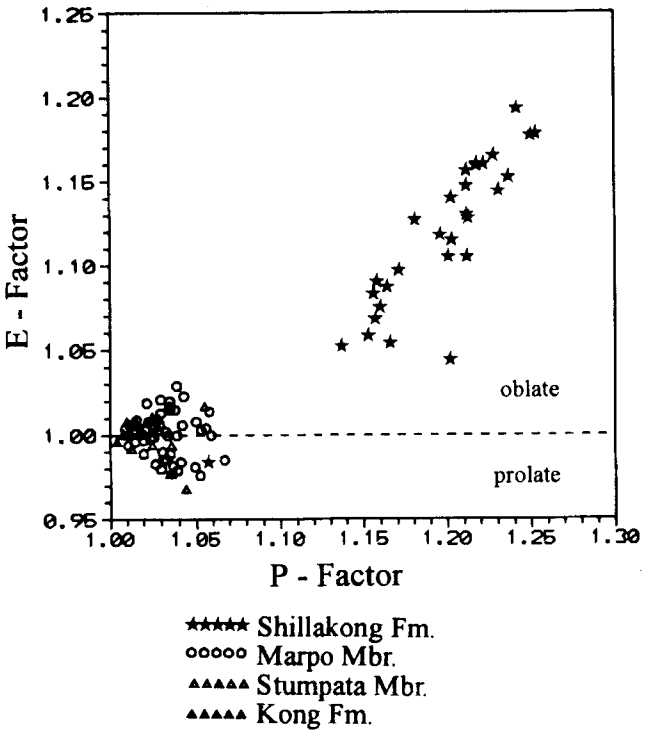


Figure 8. Left side: orthogonal vector plot showing the degmatnetization behaviour of a representative specimen from the conglomerate of the Kong Formation. Right side: specimen directions for the ChRM-A and ChRM-B components (equal area projection) in geographic coordinates.

Downloaded from https://academic.oup.com/gji/article/122/1/227/577748 by Curtin University Library user on 25 May 2026



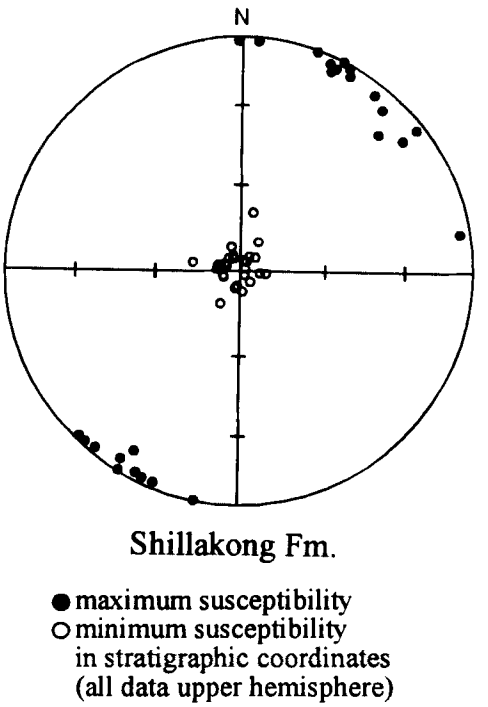


Figure 9. Top: degree of anisotropy of susceptibility (*P*-factor) plotted versus the shape of the susceptibility tensor surface (*E*-factor) for individual specimens. Bottom: directions of maximum and minimum susceptibility axes for specimens of the Shillakong Formation, plotted after tectonic correction (equal area projection).

(2) Rapid sediment accumulation of the Indus fan on the western continental margin of India about 25 Myr ago (Whiting & Karner 1991) is indicative of major uplift and cooling of the western Himalayan area.

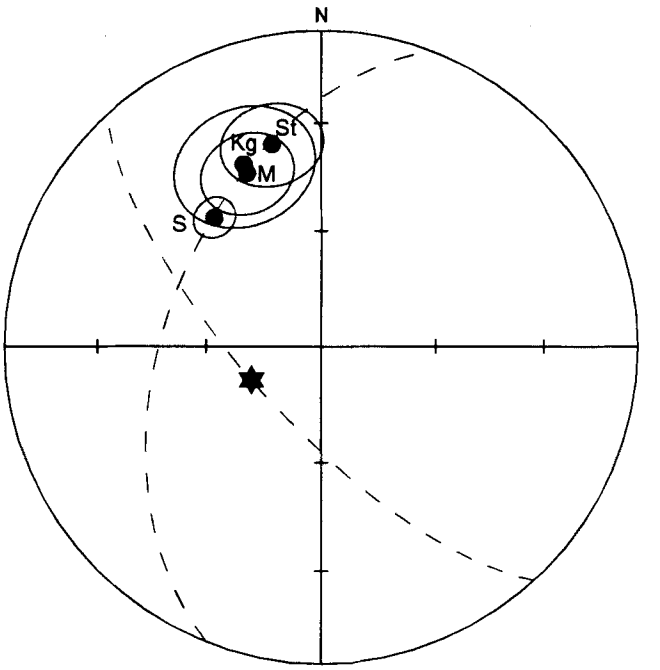


Figure 10. Site mean directions of the ChRM1 component (solid dots) with α_{95}-circles and direction of maximum susceptibility for the Shillakong Formation (solid star) in geographic coordinates (equal area projection). Two great circles are shown, defined by the remanence directions and the bedding plane, respectively.

(3) Cooling events of the High Himalayan crystallines are dated at about the Oligocene/Miocene time boundary (Le Fort 1975) or between 40 and 25 Myr (Searle & Fryer 1986), and cooling of the Tethyan sediments probably started even earlier (Le Fort 1986).

As a consequence of point (1) a calculation of the remanence acquisition age through a simple comparison of the overall mean value with the APWP of India would be certainly incorrect. From point (3) a remanence acquisition age of about 30 Myr is more likely than an age close to 40 Myr.

A comparison of pole positions of the formation mean values with other secondary remanences from the NW Himalaya (listed in Klootwijk 1984) is shown in Fig. 13. The pole positions obtained from Zanskar fit well to a swath, which is defined by other Middle–Late Tertiary secondary pole positions and which intersects the Indian APWP in the 25–45 Myr interval. The ChRM1 mean declination shows a counterclockwise rotation of $23.4 \pm 7.7°$ ($dD = \alpha_{95}/\cos I$) with respect to true north since the time of remanence acquisition (Fig. 12). Assuming a remanence acquisition age of 30 Myr a slight counterclockwise rotation of about 10° relative to the stable Indian Plate can be concluded (40 Myr/20 Myr would lead to counterclockwise rotations of about 5°/15°). This does not fit the general pattern of palaeomagnetic data from neighbouring areas within the western Himalayan syntaxis (Fig. 14). The eastern parts of the western Himalayan syntaxis show a rather consistent overall clockwise rotation of palaeodeclinations, whereas the western parts show counterclockwise rotations relative to the Indian Plate (Klootwijk *et al.* 1991). There are two

Downloaded from https://academic.oup.com/gji/article/122/1/227/577748 by Curtin University Library user on 25 May 2026

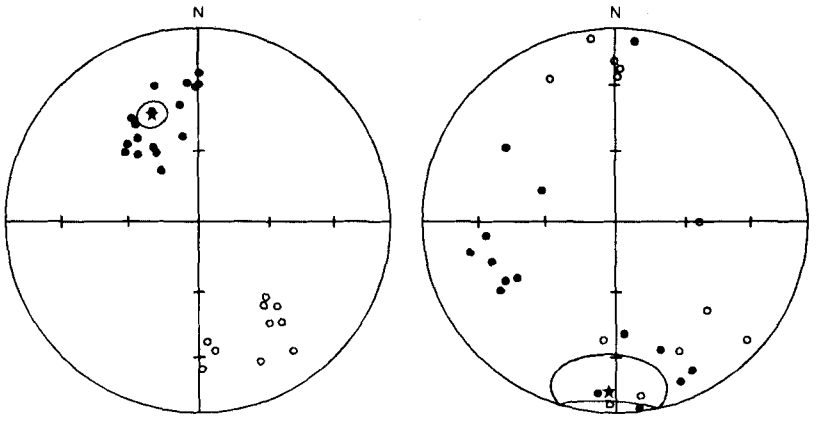


Figure 11. Site mean directions of the ChRM1 component (circles) and overall mean direction (star: mean of 27 site means) together with the α_{95}-circle before (left side) and after (right side) tectonic correction (equal area projection).

Table 5. Formation mean values for the ChRM1 component.

Mean values of the stratigraphic units

			ChRM1 (pyrrhotite component)							
stratigraphic	geologic	Ns / n	geographic corr.				stratigraphic corr.			
unit	age		Dec.	Inc.	a95	k	Dec.	Inc.	a95	k
Kong Fm.	Early Eocene	4 / 27	336.8	37.5	16.8	30.7	235.2	32.6	53.3	3.9
Stumpata Qz. Mbr.	Late Paleocene	7 / 63	346.2	34.4	11.9	26.8	338.8	−4.2	33.6	4.2
Marpo Lst. Mbr.	Late Maastrichtian	11 / 129	336.8	40.0	11.2	17.7	167.0	−6.8	17.5	7.8
Shillakong Fm.	Albian–L. Campanian	5 / 35	320.1	45.8	5.3	206.1	233.1	36.9	37.5	5.1
overall mean direction			**336.6**	**39.6**	**5.9**	**22.9**	182.4	12.2	19.1	3.1
N=Ns=27 sites (254 specimens)										
North Pole Position			**Plat. 66.5°N**		**Plong. 323.5°E**					

Ns/*n*, number of sites/specimens used for the calculation.

possible explanations for the inconsistency of our results for the Zanskar unit with Klootwijk *et al.*'s (1991) data.

(1) The age of remanence acquisition post-dates the formation of the western Himalayan syntaxis whereas the data used by Klootwijk *et al.* pre-date it. This is not very likely, however, because the formation of the western Himalayan syntaxis is expected to have occurred in Late Miocene to Pliocene times (Klootwijk *et al.* 1991), therefore post-dating the remanence acquisition age (Oligocene).

(2) Local block rotations of the Zanskar unit may have occurred, possibly caused by the obduction of the nearby Spongtang Klippe. Varying local block rotations are suggested by systematic changes of palaeoinclinations and palaeodeclinations on a NE–SW-trending profile within the investigated area (Figs 2 and 15), separating different groups of sites. Also, a correlation between tectonic strike directions and palaeodeclinations is indicated in Fig. 2. All these variations are not statistically significant, however, once the overlap of the confidence limits for different groups and the overall mean values of palaeoinclination and palaeodeclination are taken into account.

Alignments of principal tectonic features (Fig. 14) make it possible to argue that the overall mean palaeodeclination from Zanskar follows more or less this direction at NW Zanskar (about 100°), conformably with other palaeomagnetic results in the western syntaxis.

Downloaded from https://academic.oup.com/gji/article/122/1/227/577748 by Curtin University Library user on 25 May 2026

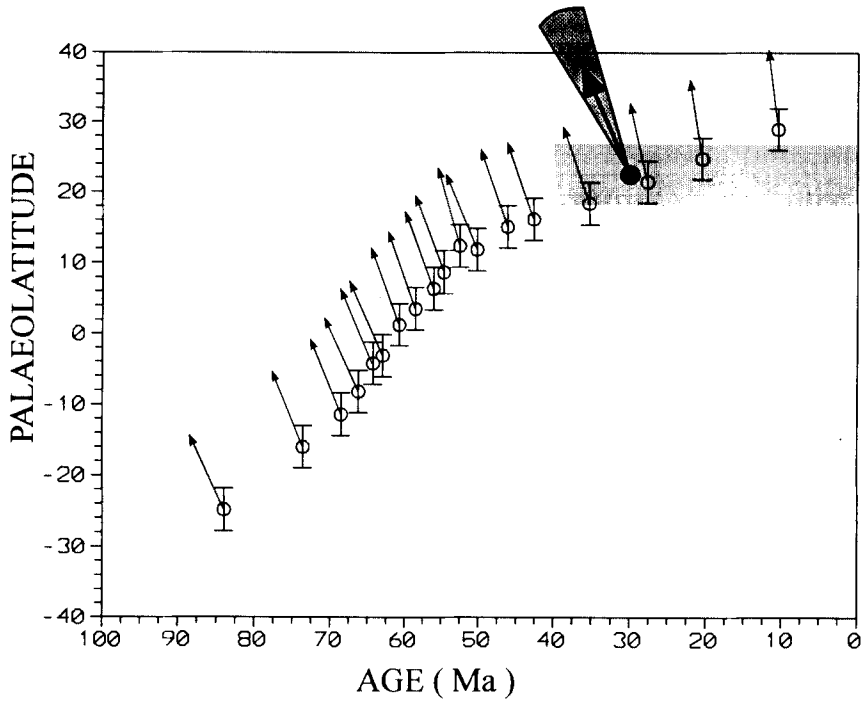


Figure 12. Expected palaeolatitudes (open circles) and palaeodeclinations (small arrows) plotted versus the geological age for the investigation area (33.9°N, 76.5°E). The expected values are calculated from the apparent polar wander path of India (according to Klootwijk *et al.* 1991), with the assumption of no relative motion between Zanskar and stable India. The solid circle and big arrow correspond to the overall mean direction of the ChRM1 component. Confidence limits for the palaeolatitude and the palaeodeclination are also shown (shaded ranges).

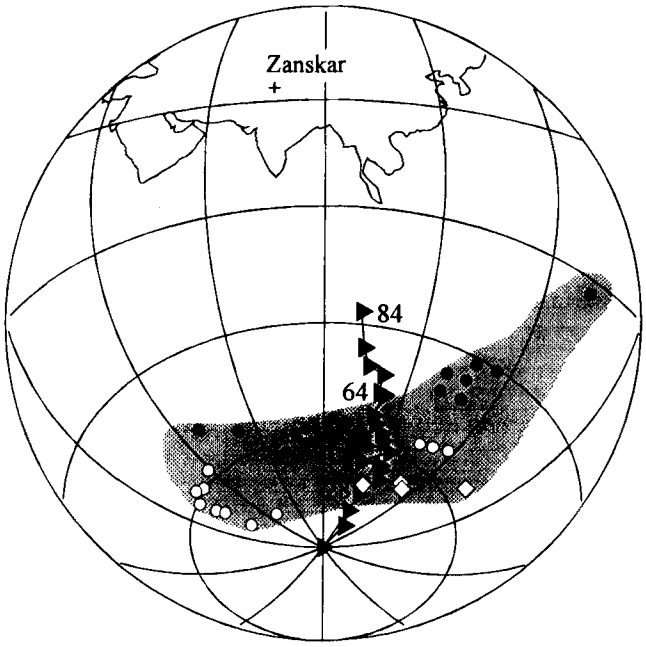


Figure 13. Comparison of the pole positions obtained from formation mean values (this paper) with other Tertiary secondary remanences from the NW Himalaya (listed in Klootwijk 1984) and an Indian APWP (Klootwijk *et al.* 1991). Black dots, Early Tertiary remanences; open dots, Early–Late Tertiary remanences; open rhombs, formation mean values, this paper, black triangles, Indian APWP, numbers indicate age in Myr. All secondary remanences together define a swath (shaded area), which intersects the Indian APWP in the 25–45 Myr interval.

Downloaded from https://academic.oup.com/gji/article/122/1/227/577748 by Curtin University Library user on 25 May 2026

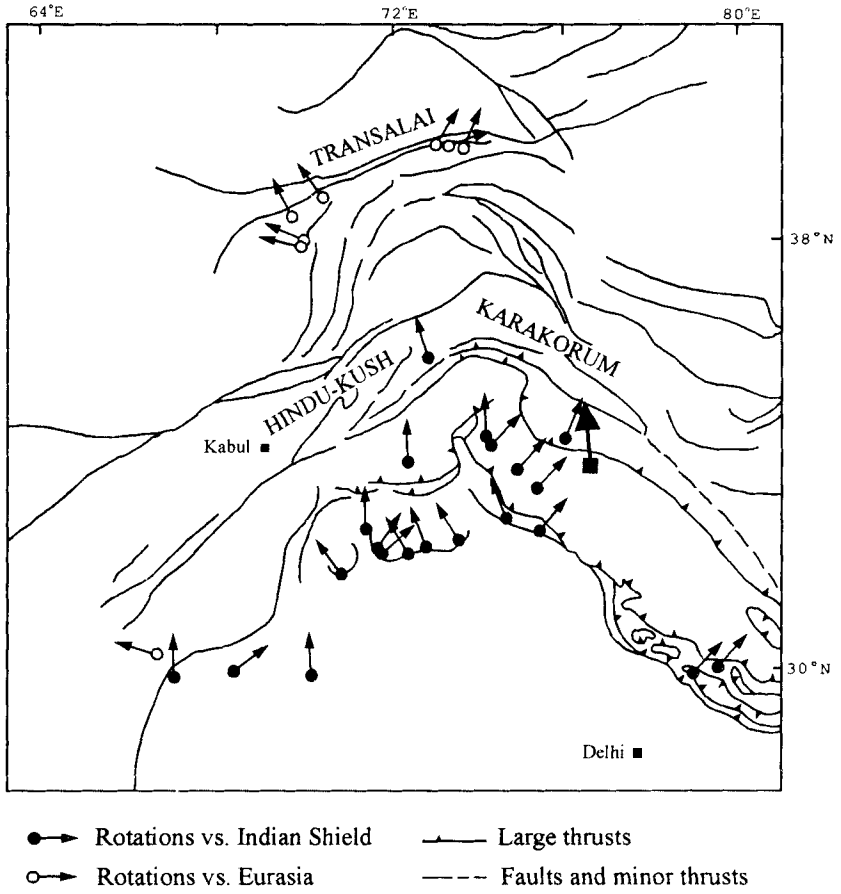


Figure 14. Rotations versus the Indian Plate deduced from palaeodeclination data within the western syntaxis of the Himalaya (Klootwijk *et al.* 1986, 1991). The ChRM1 result of this paper is shown by a square and bigger arrow.

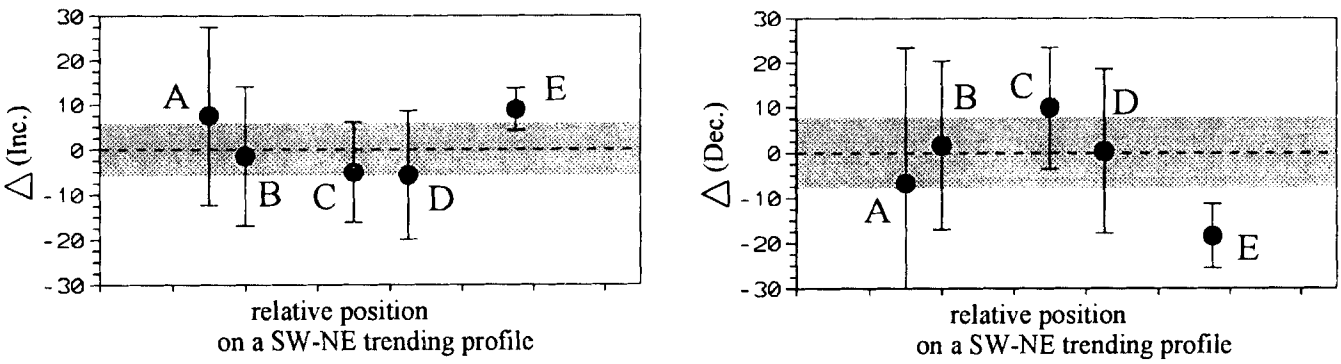


Figure 15. Inclination and declination differences between different spatial separated groups and the overall mean value. A, sites Mz3, Mz5, Mz13, STz7; B, sites M1, M2, M6, M8, M9, ST3; C, sites M17, M18, ST19, KG20, KG23, KG24, KG25; D, sites S39, M11, ST15, ST16, ST30, ST38; E, sites S44, S47, S48, S49. The confidence limits of the overall mean value are marked by the shaded area, the confidence limits of the different groups are marked by vertical bars.

ACKNOWLEDGMENTS

The authors are very much indebted to G. Fuchs, H. Willems and all the other members of the two geoscientific Zanskar expeditions (1988 and 1990) for outstanding cooperation in the field and many fruitful discussions during the evaluation of the data. The financial support of the German Research Council (DFG) is gratefully acknowledged.

REFERENCES

Appel, E. & Li Huamei, 1988. The northern edge of the Indian Plate from Albian to Paleocene, in *Himalaya–Tibet–Karakorum Workshop Meeting*, Lausanne, Abstract Book,

Downloaded from https://academic.oup.com/gji/article/122/1/227/577748 by Curtin University Library user on 25 May 2026

Musee Cantonal de Geologie, Instituts de Geologie et de Mineralogie de l'Unil, Lausanne, Switzerland, 10 p.

Appel, E., Müller, R. & Widder, R.W., 1991. Palaeomagnetic results from the Tibetan Sedimentary Series of the Manang area (north central Nepal), *Geophys. J. Int.*, **104,** 255–266.

Besse, J. & Courtillot, V., 1988. Paleogeographic maps of the Indian Ocean bordering continents since the Upper Jurassic, *J. geophys. Res.*, **93,** 11 791–11 808.

Besse, J., Courtillot, V., Pozzi, J.P., Westphal, M. & Zhou, Y.K., 1984. Palaeomagnetic estimates of crustal shortening in the Himalayan thrusts and Zangpo suture, *Nature*, **311,** 621–626.

Coward, M.P., Windley, B.F., Broughton, R.D., Luff, I.W., Petterson, M.G., Pudsey, C.J., Rex, D.C. & Asif Khan, M., 1986. Collision tectonics in the NW Himalayas, in *Collision Tectonics*, Geol. Soc. Spec. Publ., No. 19, pp. 203–219, eds Coward, M.P. & Ries, A.C., Blackwell Scientific Publications, Oxford.

Dewey, J.F., Cande, S. & Pitman, W.C., 1989. Tectonic evolution of the India/Eurasia collision zone, *Eclog. geol. Helv.*, **82,** 717–734.

Fisher, R.A., 1953. Dispersion on a sphere, *Proc. R. Soc. Lond.*, A, **217,** 295–305.

Fuchs, G., 1982a. (Explanations of the) *Geological-tectonic map of the Himalaya*, 1:2.000.000, Geol. Bundesanstalt, Wien.

Fuchs, G., 1982b. The geology of western Zanskar, *Jahrb. Geol. Bundesanstalt, Wien*, **125,** 1–50.

Fuchs, G., 1987. The geology of the Southern Zanskar (Ladakh)—Evidence for the autochthony of the Tethys Zone of the Himalaya, *Jahrb. Geol. Bundesanstalt, Wien*, **130,** 465–491.

Fuchs, G. & Willems, H., 1990. The final stages of sedimentation in the Tethyan Zone of Zanskar and their geodynamic significance (Ladakh-Himalaya), *Jahrb. Geol. Bundesanstalt, Wien*, **133,** 259–273.

Gaetani, M. & Garzanti, E., 1991. Multicyclic history of the Northern India continental margin (Northwestern Himalaya), *Am. Assoc. Petrol. Geol. Bull.*, **75,** 1427–1446.

Gaetani, M., Nicora, A., Premoli Silva, I., Fois, E. & Garzanti, E., 1983. Upper Cretaceous and Paleocene in Zanskar Range (NW Himalaya), *Riv. Ital. Paleont.*, **89,** 81–118.

Gansser, A., 1964. *The Geology of the Himalayas*, New York, Wiley, 289p.

Garzanti, E. & Brignoli, G., 1989. Low temperature metamorphism in the Zanskar sedimentary nappes (NW Himalaya, India), *Eclog. Geol. Helv.*, **82,** 669–684.

Jaeger, J.J., Courtillot, V. & Tapponnier, P., 1989. Paleontological view of the ages of the Deccan Traps, the Cretaceous/Tertiary boundary and the India–Asia collision, *Geology*, **17,** 316–319.

Kirschvink, J.L., 1980. The least-square line and plane and the analysis of palaeomagnetic data, *Geophys. J.R. astr. Soc.*, **62,** 699–718.

Klootwijk, C.T., 1984. A review of Indian Phanerozoic Palaeomagnetism: Implications for the India–Asia collision, *Tectonophysics*, **105,** 331–353.

Klootwijk, C.T. & Bingham, D.K., 1980. The extent of Greater India, III. Palaeomagnetic data from the Tibetan Sedimentary Series, Thakkola region, Nepal Himalaya, *Earth planet. Sci. Lett.*, **51,** 381–405.

Klootwijk, C.T., Conaghan, P.J. & Powell, C.McA., 1985. The Himalayan arc: large-scale continental subduction, oroclinal bending and back-arc spreading, *Earth planet. Sci. Lett.*, **75,** 167–183.

Klootwijk, C.T., Gee, J.S., Peirce, J.W. & Smith, G.M., 1991. Constraints on the India–Asia Convergence: Paleomagnetic Results from Ninetyeast Ridge, in *Proc. ODP, Sci. Results*, No. 121, pp. 777–881, eds Weissel, J., Peirce, J., Taylor, E. & Alt, J.

Klootwijk, C.T., Sharma, M.L., Gergan, J., Shah, S.K. & Gupta, B.K., 1986. Rotational overthrusting of the northwestern Himalaya: further palaeomagnetic evidence from the Riasi thrust sheet, Jammu foothills, India, *Earth planet. Sci. Lett.*, **80,** 375–393.

Klootwijk, C.T., Shah, S.K., Gergan, J., Sharma, M.L., Tirkey, B. & Gupta, B.K., 1983. A palaeomagnetic reconnaissance of Kashmir, northwestern Himalaya, India, *Earth planet. Sci. Lett.*, **63,** 305–324.

Le Fort, P., 1975. Himalayas: the collided range. Present knowledge of the continental arc, *Am. J. Sci.*, **275A,** 1–44.

Le Fort, P., 1986. Metamorphism and magmatism during the Himalayan collision, in *Collision Tectonics*, Geol. Soc. Spec. Publ., No. 19, pp. 159–172, eds Coward, M.P. & Ries, A.C., Blackwell Scientific Publications, Oxford.

McElhinny, M.W., 1964. Statistical significance of the fold test in palaeomagnetism, *Geophys. J.R. astr. Soc.*, **8,** 338–340.

Patriat, P. & Achache, J., 1984. India–Eurasia collision chronology has implications for crustal shortening and driving mechanism of plates, *Nature*, **311,** 615–621.

Patzelt, A., Li Huamei, Wang Junda & Appel, E., 1995. Palaeomagnetism of Cretaceous to Tertiary formations from southern Tibet: Evidence for the extent of the northern margin of India prior to the collision with Eurasia, *Tectonophysics*, submitted.

Powell, C.McA., 1986. Continental underplating model for the rise of the Tibetan Plateau, *Earth planet. Sci. Lett.*, **81,** 79–94.

Rochette, P., 1988. La susceptibilité anisotrope des roches faiblement magnétiques, origines et applications, *PhD thesis*, University of Grenoble.

Rochette, P., Scaillet, B., Guillot, S., LeFort, P. & Pêcher, A., 1994. Magnetic properties of the High Himalayan leucogranites: structural implications, *Earth planet. Sci. Lett.*, **126,** 217–234.

Searle, M.P. & Fryer, B.J., 1986. Garnet, tourmaline and muscovite bearing leucogranites, gneisses and migmatites of the Higher Himalayas from Zanskar, Kulu, Lahoul and Kashmir, in *Collision Tectonics*, Geol. Soc. Spec. Publ., No. 19, pp. 185–201, eds Coward, M.P. & Ries, A.C., Blackwell Scientific Publication, Oxford.

Tonarini, S., Villa, L., Oberli, F., Meier, M., Spencer, D.A., Pognante, U. & Ramsay, J.G., 1993. Eocene age of eclogite metamorphism in Pakistan Himalaya: implications for India–Eurasian collision, *Terra Nova*, **5,** 13–20.

Whiting, B.M. & Karner, G.D., 1991. 25Ma flexural modification of western Indian continental margin: effects of Indus fan loading, *EOS, Trans. Am. geophys. Un.*, **72,** 472.

Zhu, Z. & Teng, J., 1984. Preuves paléomagnetiques de la dérive vers le Nord de fragments de la plaque Indienne aprés la séparation du continent de Gondwana et de leur collision avec la plaque Eurasiatique, in *Mission Françoise-Chinoise au Tibet 1980*, pp. 15–20, eds Mercier, J.L. & Li Guangcen, CNRS, Paris.

Downloaded from https://academic.oup.com/gji/article/122/1/227/577748 by Curtin University Library user on 25 May 2026